

The Cosmic Dipole Anomaly as Regime-Dependent Anisotropy in Energy-Flow Cosmology

Morten Magnusson

Independent Researcher, Sola, Norway

ORCID: 0009-0002-4860-5095

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Abstract. Recent observations of the galaxy number count dipole show an amplitude 2–3 times larger than predicted by the kinematic interpretation within Λ CDM. This working note examines whether Energy-Flow Cosmology (EFC) contains a natural mechanism for excess large-scale anisotropy. We identify the key structural ingredient: in EFC, isotropy is a regime condition valid within $L0/L1$, not an absolute postulate. The entropy field $S(\mathbf{x}, z)$ need not be spatially uniform at super-Hubble scales, and any residual gradient $\nabla S|_{\text{LSS}}$ imprints a preferred direction in structure growth via the regime-dependent coupling $\mu(S)$. We derive the leading-order dipole contribution, state the observational predictions, and identify the open gaps that prevent empirical closure. This is a working note establishing the logical connection; it is not a detection claim.

1 The Observational Anomaly

The cosmic dipole in the CMB is well explained by our peculiar velocity $v \approx 370$ km/s relative to the rest frame. Under the kinematic hypothesis, the same velocity should produce a dipole in galaxy number counts of amplitude $\mathcal{D}_{\text{kin}} \approx 2v/c \approx 0.0045$ (including the combined effect of aberration and Doppler boosting) [1,2].

Multiple independent analyses—using NVSS radio sources, CatWISE2020 quasars, and WISE-SuperCOSMOS galaxies—consistently find a dipole amplitude $\mathcal{D}_{\text{obs}} \approx 0.008\text{--}0.015$, roughly 2–3 times larger than kinematic predictions [3,4]. The excess is approximately aligned with the CMB dipole direction but significantly exceeds the expected magnitude.

Within Λ CDM, the cosmological principle requires statistical isotropy on large scales. The excess dipole therefore constitutes either a systematic artifact, a statistical fluctuation, or a signal of physics beyond the standard model.

2 EFC Structural Ingredients

2.1 Isotropy as regime condition

In the GRC synthesis [5], isotropy is treated as a *regime condition*, not an absolute uniqueness postulate. Specifically:

- In L0 (CMB regime): the entropy field S is nearly uniform ($S \rightarrow S_{\text{max}}$, thermal equilibrium), and isotropy holds to the precision of CMB observations ($\delta T/T \sim 10^{-5}$).
- In L1 (cosmological background): $S(z)$ evolves with redshift but is assumed spatially homogeneous in the background Friedmann equation.
- In L2 (structure-dominated): $S(\mathbf{x})$ varies spatially, breaking isotropy locally. This is where galaxies, clusters, and voids live.

The critical observation is that the transition from L1 to L2 need not be spatially uniform. If structure formation proceeded slightly faster in one direction—due to initial conditions, topology, or pre-inflationary gradients—then $S(\mathbf{x}, z)$ inherits a large-scale gradient.

2.2 The entropy gradient mechanism

In EFC, the effective gravitational coupling depends on the entropy field:

$$G_{\text{eff}}(S) = G_N(1 + \mu(S)), \quad (1)$$

where $\mu(S) = \mu_0 \cdot f(S)$ with $\mu_0 = 0.415$ from SPARC calibration [6]. The function $f(S)$ is monotonically decreasing: lower entropy (more structure) gives stronger effective gravity.

If S has a large-scale gradient, then G_{eff} has a directional dependence:

$$\Delta G_{\text{eff}}(\hat{n}) \approx G_N \mu_0 f'(\bar{S}) (\nabla S \cdot \hat{n}) L_{\text{grad}}, \quad (2)$$

where $\bar{S}$ is the volume-averaged entropy, f' is the derivative of the enhancement function, $\hat{n}$ is a direction on the sky, and L_{grad} is the coherence length of the gradient.

3 Dipole Prediction

3.1 Leading-order dipole from ∇S

The galaxy number count dipole receives a contribution from direction-dependent structure growth. In a region where G_{eff} is enhanced, structure grows faster, producing more observable galaxies above a flux threshold.

The fractional excess in galaxy counts along direction $\hat{n}$ is:

$$\frac{\delta N}{N}(\hat{n}) \approx b_{\text{eff}} \cdot \frac{\partial \ln \sigma_8}{\partial \ln G_{\text{eff}}} \cdot \frac{\Delta G_{\text{eff}}(\hat{n})}{G_N}, \quad (3)$$

where b_{eff} is the effective bias of the tracer population and $\sigma_8 \propto G_{\text{eff}}^{1/2}$ in the linear growth approximation.

Combining Eqs. (2) and (3), the EFC-induced dipole amplitude is:

$$\mathcal{D}_{\text{EFC}} \approx \frac{b_{\text{eff}}}{2} \mu_0 |f'(\bar{S})| \frac{|\nabla S| L_{\text{grad}}}{\bar{S}}. \quad (4)$$

3.2 Order-of-magnitude estimate

For the observed excess $\mathcal{D}_{\text{excess}} = \mathcal{D}_{\text{obs}} - \mathcal{D}_{\text{kin}} \approx 0.005\text{--}0.010$:

Using $b_{\text{eff}} \approx 1.5$ (typical for radio/IR-selected samples), $\mu_0 = 0.415$, and $|f'(\bar{S})| \sim 1$ near the regime transition:

$$\frac{|\nabla S| L_{\text{grad}}}{\bar{S}} \approx \frac{2 \mathcal{D}_{\text{excess}}}{b_{\text{eff}} \mu_0} \approx \frac{2 \times 0.007}{1.5 \times 0.415} \approx 0.023. \quad (5)$$

This requires a $\sim 2\%$ variation in the entropy field across the observable volume—a small but non-zero departure from perfect homogeneity, consistent with the known $\sim 10^{-5}$ CMB anisotropy growing by a factor $\sim 10^3$ through structure formation.

4 Testable Predictions

- (P1) **Redshift dependence.** The EFC dipole excess should grow with decreasing redshift (as S decreases and $\mu(S)$ increases). The kinematic dipole is redshift-independent. *Discriminator:* measure the dipole amplitude in tomographic redshift bins.
- (P2) **Correlation with S_8 anisotropy.** If the dipole is driven by ∇S , then the growth rate $f\sigma_8$ should show a corresponding directional dependence at the $\sim 2\%$ level. *Test:* directional $f\sigma_8$ from DESI/Euclid.
- (P3) **Alignment with structure asymmetry.** The EFC dipole direction should correlate with independently measured bulk flow directions and the CMB dipole, but the *amplitude* should exceed the kinematic prediction. This is what is observed.
- (P4) **Scale dependence.**
The EFC mechanism operates at super-cluster scales ($k < 0.01 \text{ Mpc}^{-1}$), where the stiffness response $R(k) \propto k^{-4}$ becomes significant. The excess dipole should be more pronounced for tracers sampling large volumes.

5 Open Gaps

- (G1) **Origin of $\nabla S|_{\text{LSS}}$.** This note establishes that a large-scale entropy gradient *can* produce the observed dipole excess within EFC. It does not derive the gradient from initial conditions. A complete account requires connecting $\nabla S|_{\text{LSS}}$ to inflationary/pre-inflationary physics or to the topology of the grid substrate.
Speculative candidate: In EFC, the entropy field S is sourced by structure formation via $\Gamma(\rho)$. The cosmic web is not isotropic at $\lesssim 300$ Mpc scales—it contains super-void/supercluster asymmetries that imprint a net ∇S on the last-scattering surface. The required $\sim 2\%$ gradient could arise from the known $\sim 10\%$ matter density asymmetry in the local supercluster environment, amplified through the non-linear $\mu(S)$ response. This is not derived here but constitutes a concrete physical scenario amenable to N-body estimation.
- (G2) **Quantitative $f(S)$ profile.** The estimate in Eq. (5) uses $|f'| \sim 1$. The actual profile $f(S)$ is constrained by SPARC at L2 scales but not at the super-Hubble coherence scales relevant for the dipole. This requires extrapolation beyond current calibration.
- (G3) **Separation from kinematic contribution.** The observed dipole is a sum of kinematic and intrinsic (structure) components. Isolating the EFC contribution requires multi-tracer, multi-redshift analysis that has not been performed within this framework.
- (G4) **Consistency with CMB isotropy.** The CMB constrains anisotropy at $z \approx 1100$ to $\sim 10^{-5}$. The $\sim 2\%$ entropy gradient at $z \approx 0$ must be consistent with this constraint under EFC's $S(z)$ evolution. This is plausible (growth factor $\sim 10^3$) but requires explicit verification.

6 Summary

EFC contains a natural mechanism for excess large-scale anisotropy: the entropy field $S(\mathbf{x}, z)$ need not be spatially uniform, and any large-scale gradient directly modulates structure growth via $G_{\text{eff}}(S)$. The required entropy gradient ($\sim 2\%$ across the observable volume) is consistent with known CMB constraints under standard growth amplification.

This note establishes logical consistency, not detection. The dipole anomaly is *compatible* with EFC’s regime-dependent isotropy framework. Empirical closure requires predictions P1–P4 to be tested against forthcoming DESI/Euclid tomographic data.

Status: Working note. Gap G1 (origin of gradient) and G2 (quantitative $f(S)$ profile) are open.

References

- [1] G. F. R. Ellis and J. E. Baldwin, “On the expected anisotropy of radio source counts,” MNRAS **206**, 377 (1984).
- [2] C. Blake and J. Wall, “A velocity dipole in galaxy number counts,” Nature **416**, 150 (2002).
- [3] N. J. Secrest *et al.*, “A Test of the Cosmological Principle with Quasars,” ApJL **908**, L51 (2021). DOI: [10.3847/2041-8213/abdd40](https://doi.org/10.3847/2041-8213/abdd40).
- [4] T. M. Dam, C. A. P. Bengaly, and R. Maartens, “Testing the cosmological principle with CatWISE quasars: a Bayesian analysis of the number-count dipole,” JCAP **2023**, 062 (2023).
- [5] M. Magnusson, “Grid-Rendering Cosmology: Synthesis Paper,” EFC Paper Series (2026). Integration Guide v2.0.
- [6] M. Magnusson, “EFC Phase 3 SPARC Validation,” DOI: [10.6084/m9.figshare.31224538](https://doi.org/10.6084/m9.figshare.31224538) (2026).
- [7] M. Magnusson, “EFC Regime-Transition Fit to DESI DR2 BAO,” DOI: [10.6084/m9.figshare.31230703](https://doi.org/10.6084/m9.figshare.31230703) (2026).
- [8] M. Magnusson, “Systematic Localization of Late-Time Cosmological Signals in Modified Gravity,” DOI: [10.6084/m9.figshare.31368433](https://doi.org/10.6084/m9.figshare.31368433) (2026).